

Math 521, Section 2
Homework 12

Problem 1. Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \rightarrow Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Problem 2. Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \rightarrow Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Problem 3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function $f(x) = \frac{1}{1+x^2}$.

- (a) Prove that f is continuous.
- (b) Find an open set $U \subseteq \mathbb{R}$ so that $f(U)$ is not open.
- (c) Find a closed set $F \subseteq \mathbb{R}$ so that $f(F)$ is not closed.
- (d) Find a set $A \subseteq \mathbb{R}$ so that $f(\overline{A}) \neq \overline{f(A)}$.

Problem 4. Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

- (a) Prove that the functions $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)
- (b) Prove that if $h : [0, 1] \rightarrow \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y -axis.

Problem 5. Going forward, you may assume (no proof necessary) any fact about $\sin x$ and $\cos x$ from calculus, such as $\sin, \cos : \mathbb{R} \rightarrow [-1, 1]$ are continuous functions.

Prove that the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum. (Hint: Notice that we cannot directly apply the extreme value theorem here. However, we can apply it to any compact subset of $\mathbb{R}^2$.)

Problem 6. Let X and Y be metric spaces and $f : X \rightarrow Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \rightarrow X$ is also continuous. (Hint: It suffices to show that the preimage of any closed subset of X under f^{-1} is closed in Y .)